Rapid decay survives exponential distortion

A nondegenerate amalgam answering the final open consideration in arXiv:2509.08754

Automated mathematics discovery run `fa_banach_001`
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Status: candidate full solution, likely valid. We construct a nondegenerate amalgamated free product whose universal length has rapid decay although one vertex length is exponentially distorted. The construction uses countable groups with weighted lengths and does not address the neighboring finitely generated word-length questions.

1 Source question

Kunnawalkam Elayavalli, Patchell and Teryoshin develop subexponential decay (SD), prove permanence theorems for amalgamated free products, and quantify the effect of distortion [1]. They close their introduction with three open directions. The last asks whether there are systematic or specific instances in which RD or SD survives high distortion:

> To conclude the introduction we include certain open considerations of interest. A natural question we are unable to address, in light of Theorem 2.13 is if there is a finitely generated group G with no super-polynomial growth amenable subgroups which has SD with respect to the word length, but not RD. It would also be interesting if finitely presented such examples exist. Secondly, regarding selflessness, it is proved in [1] that any countable selfless group (which is an effective mixed identity freeness property of a group) with RD satisfies that the reduced group C^*-algebra is selfless in the sense of Robert. It is very natural to now ask if one can replace RD with SD in the above. As we explained above, the proof in [1], cannot be generalized to this case as such. Regarding general amalgamated free products, it is a natural question if there can be any systematic or specific instances where RD/SD is still preserved, despite having high distortion.

We answer this final question affirmatively with a fully explicit example. The setting is exactly the source's general framework of countable groups equipped with integer-valued length functions.

2 Construction and result

Let

$$A = \bigoplus_{n\geq 0} C_2$$

and denote its standard basis by $e_0, e_1, \dots$. Every $a \in A$ has a unique finite support $S(a) \subset \mathbb{N} \cup \{0\}$. Define two weighted lengths on A by

$$\ell_0(a) = \sum_{n \in S(a)} 2^n, \qquad \ell_1(a) = \sum_{n \in S(a)} 2^{2^n}. \tag{1}$$

For $u \in C_k$, put $\delta_k(u) = 0$ if $u = 1$ and $\delta_k(u) = 1$ otherwise. Set

$$\begin{aligned} G &= A \times C_2, & L_G(a,u) &= \ell_1(a) + \delta_2(u), \\ H &= A \times C_3, & L_H(a,v) &= \ell_0(a) + \delta_3(v). \end{aligned} \tag{2}$$

We identify A with $A \times \{1\}$ in both factors and form $\Gamma = G *_A H$. Both inclusions of A are proper.

Let L^U be the universal length on Γ defined in [1]: in a factorization by elements of $G \cup H$, each letter is charged its vertex length, while a letter in A may be charged either restriction.

Theorem 1 (RD beyond the polynomial-distortion regime)**.** *There is a canonical isomorphism*

$$\Gamma \cong A \times (C_2 * C_3). \tag{3}$$

If $|q|$ denotes the standard free-product word length on $C_2 * C_3$, then

$$L^U(a,q) = \ell_0(a) + |q|. \tag{4}$$

The pair (Γ, L^U) has rapid decay.

On the other hand, define the distortion function of the first vertex by

$$D(r) = \max\{L_G(g) : g \in G,\ L^U(g) \le r\}.$$

For every integer $r \ge 1$,

$$2^{r/2} < D(r) \le 2^{r+1} + 1. \tag{5}$$

Thus $(L^U|_G, L_G)$ is exponentially distorted. In particular, the polynomial-distortion hypothesis in the source's RD permanence theorem fails.

3 Proof intuition

The Boolean group is assigned two radically different proper geometries. Under the light weights 2^n, binary expansion gives exactly one group element of every nonnegative integer length, so balls have linear size. The heavy weights 2^{2^n} are exponentially larger but only make balls smaller.

The amalgam is transparent because A is central in both vertices: it factors as A times the virtually free group $C_2 * C_3$. In the universal length, an A coordinate can always be moved to the lighter vertex, so the light length wins exactly. This gives RD by direct-product permanence. Seen from the heavy vertex, however, the binary element of light length r has heavy length at least $2^{r/2}$.

4 Proof of Theorem 1

4.1 Lengths and the amalgam

The two functions in (1) are weighted word lengths. Indeed, addition in A takes symmetric differences of supports, so cancellation can only decrease the sum of positive weights. Hence L_G and L_H are length functions.

The group A is central in both G and H. The universal property of the amalgam gives mutually inverse homomorphisms between $G *_A H$ and $A \times (C_2 * C_3)$, proving (3).

4.2 Exact computation of the universal length

The upper bound in (4) is immediate: represent a as an element of the H copy, at cost $\ell_0(a)$, and spell the reduced word q, at cost $|q|$.

For the reverse bound, consider any factorization of (a, q) by letters $x_i \in G \cup H$. Write the A coordinate of x_i as a_i and its finite cyclic coordinate as c_i. Every allowed charge for x_i is at least

$$\ell_0(a_i) + \delta(c_i),$$

because $\ell_1 \geq \ell_0$. Therefore the total charge is at least

$$\sum_i \ell_0(a_i) + \sum_i \delta(c_i) \geq \ell_0(a) + |q|.$$

The last inequality uses the triangle inequality first in A and then for the word length in $C_2 * C_3$. Taking the infimum over all factorizations proves (4). Notice that this also rules out savings obtained by bundling an A coordinate and a cyclic coordinate into the same factor letter.

4.3 Rapid decay

The map

$$a \longmapsto \ell_0(a) = \sum_{n \in S(a)} 2^n$$

is the binary-expansion bijection from A to the nonnegative integers. Hence

$$|B_{\ell_0}(R)| = R + 1 \qquad (R \in \mathbb{N} \cup \{0\}). \tag{6}$$

If $\varphi \in \mathbb{C}[A]$ is supported in this ball, then

$$\|\varphi\|_{C_r^*(A)} \leq \|\varphi\|_1 \leq \sqrt{R+1}\, \|\varphi\|_2.$$

Thus (A, ℓ_0) has RD. Since $\ell_1 \geq \ell_0$, its balls are smaller and the same polynomial estimate proves RD for (A, ℓ_1). The finite cyclic coordinates in (2) only multiply ball cardinalities by fixed constants, so both vertex groups have RD.

The source's free-product Lemma 2.12 shows that $C_2 * C_3$, with $|\cdot|$, has RD: finite groups have constant decay functions, and the lemma multiplies the maximum input function by the polynomial $(2R+1)^{7/2}$. Its direct-product Lemma 2.11 multiplies decay functions. Applying it to (A, ℓ_0) and $(C_2 * C_3, |\cdot|)$, and using (4), proves RD for (Γ, L^U).

4.4 Exponential distortion

For each integer $r \geq 1$, let $a_r \in A$ be the element whose support is the binary expansion of r. Then $\ell_0(a_r) = r$. If 2^N is its leading binary digit, then $2^N > r/2$, so

$$L_G(a_r) = \ell_1(a_r) \geq 2^{2^N} > 2^{r/2}. \tag{7}$$

This proves the lower bound in (5).

Conversely, suppose $g = (a, u) \in G$ and $L^U(g) \leq r$. If 2^N is the largest light weight in $S(a)$, then $2^N \leq \ell_0(a) \leq r$. The superincreasing heavy weights satisfy

$$\ell_1(a) \leq \sum_{n=0}^{N} 2^{2^n} \leq 2\, 2^{2^N} \leq 2^{r+1}.$$

Adding $\delta_2(u) \leq 1$ proves the upper bound in (5) and completes the proof.

5 Scope and novelty check

This is a full affirmative answer to the final request for a specific high-distortion amalgam preserving RD/SD. It works in the paper's stated countable-group framework and is nondegenerate, since A has indices two and three in the vertex groups. The example deliberately exploits weighted lengths. It does not answer the separate preceding questions about finitely generated or finitely presented examples with word length.

The run's cheap indexes were searched by arXiv identifier, exact title, and the core rapid-decay, distortion and amalgam phrases. A bounded primary arXiv search found the source paper but no later resolution or occurrence of this weighted Boolean construction. This is not an exhaustive novelty certification.

References

[1] S. Kunnawalkam Elayavalli, G. Patchell and L. Teryoshin, *Some remarks on decay in countable groups and amalgamated free products*, arXiv:2509.08754 (2025).